4 Fair Division

4.3 Sealed Bids Method

Definition 4.3.1 The method begins by compiling a list of items to be divided. Then:

1. Each party involved lists, in secret, a dollar amount they value each item to be worth. This is their sealed bid.

2. The bids are collected. For each party, the value of all the items is totaled, and divided by the number of parties. This defines their fair share.

3. Each item is awarded to the highest bidder.

4. For each party, the value of all items received is totaled. If the value is more than that party's fair share, they pay the difference into a holding pile. If the value is less than that party 's fair share, they receive the difference from the holding pile. This ends the initial allocation.

5. In most cases, there will be a surplus, or leftover money, in the holding pile. The surplus is divided evenly between all the players. This produces the final allocation.

Example 4.3.2 Sam and Omar have cohabitated for the last 3 years, during which time they shared the expense of purchasing several items for their home. Sam has accepted a job in another city, and now they find themselves needing to divide their shared assets. Each records their bids for each item as shown below.

	Sam	Omar
Couch	\$150	\$100
TV	\$200	\$250
Game Console	\$250	\$150
Sound System	\$50	\$100

Example 4.3.3 Jamal, Maggie, and Kendra are dividing an estate consisting of a house, a vacation home, and a small business. Their valuations (in thousands) are shown below. Determine the final allocation.

	Jamal	Maggie	Kendra
House	\$250	\$300	\$280
Vacation Home	\$170	\$180	\$200
Small Business	\$300	\$255	\$270

Example 4.3.4 As part of an inheritance, four children, Abby, Ben and Carla, are dividing four vehicles using Sealed Bids. Their bids (in thousands of dollars) for each item is shown below. Find the final allocation.

	Abby	Ben	Carla
Motorcycle	10	9	8
Car	10	11	9
Tractor	4	1	2
Boat	7	6	4

Example 4.3.5 Fair division does not always have to be used for items of value. It can also be used to divide undesirable items. Suppose Chelsea and Mariah are sharing an apartment, and need to split the chores for the household. They list the chores, assigning a negative dollar value to each item; in other words, the amount they would pay for someone else to do the chore (a per week amount). We will assume, however, that they are committed to doing all the chores themselves and not hiring a maid. Find a fair division if the bids are given as shown below.

	Chelsea	Mariah
Vacuuming	-$10	-$8
Cleaning bathroom	-$14	-$20
Doing Dishes	-$4	-$6
Dusting	-$6	-$4